Krishnaa Vadivel | Computational Physics | Distributed Computing | Machine Learning

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Summary

Computational physics researcher building distributed scientific software and using machine learning to accelerate calculations for first-principles materials research. Experience spans Python/C++, CUDA/ROCm, MPI/OpenMP, Linux-based HPC environments, and Docker containers for packing applications across environments.

Research Experience

Yale University, Electrical Engineering
PhD Research Assistant — *2021–present*

- Developed distributed simulation software in Python and C++ for first-principles workloads as part of an INCITE award using exascale resources, with MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier at OLCF, Perlmutter at LBNL, and related national-lab clusters.
- Formulated coherent-state and path-integral viewpoints for exciton-phonon dynamics in self-trapped excitons and excitonic polarons, connecting DFT, DFPT, and GW/BSE-style methods.
- Used PyTorch, PyTorch Geometric, and equivariant graph neural networks to accelerate self-trapped-exciton calculations and related materials simulations.
- Packaged scientific applications in Docker containers for deployment across different computing environments and used distributed debuggers such as TotalView and Linaro DDT to track down bugs on HPC systems.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS research in 2024, 2025, and 2026.

Industry Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer — *2019–2021*

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built full-stack server applications with C#, ASP.NET, and Microsoft SQL Server for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight
Photonics Technical Writer — *2018–2019*

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Exciton-Phonon HPC Research: Developed distributed first-principles calculations for self-trapped excitons, many-body theory, and scalable materials simulations.

Equivariant Scientific ML: Developed PyTorch Geometric equivariant models to accelerate molecular and excitonic calculations.

Meep Ring Resonator FDTD: Designed a nanophotonic ring-resonator band-pass filter in telecom bands using

Meep-based FDTD simulation.

dolfinx Finite-Element Analysis: Applied dolfinx-based finite-element analysis to frequency studies of steel pipes in well environments.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, SciPy, SymPy, matplotlib, dolfinx

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Platforms: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Perlmutter

Domains: First-principles materials modeling, many-body theory, distributed computing, computational physics, numerical PDEs